

STATA GROUP ASSIGNMENT 1

Operating Profitability and Portfolio Excess Returns

Value-weighted deciles | July 1963–July 2026 | 757 months

1 Data and portfolio construction

We use Kenneth French's ten value-weighted portfolios formed on operating profitability (OP), ordered from the lowest-profitability decile (1) to the highest (10). Portfolios are formed each June using NYSE breakpoints. OP is prior-year revenue less cost of goods sold, interest expense, and selling, general and administrative expenses, divided by book equity.

Monthly returns are matched by calendar month to the risk-free return (RF) in the Fama/French three-factor file. The common sample contains 757 complete observations per decile. The files were downloaded on September 15, 2026 and use the July 2026 CRSP vintage; only the monthly value-weighted return block is selected.

2 Excess returns and inference

For portfolio d , monthly excess return is $x_{d,t} = R_{d,t} - R_{f,t}$, measured in percentage points. We test $H_0 : \mu_d = 0$ against $H_1 : \mu_d \neq 0$ using

$$t_d = \frac{\bar{x}_d}{s_d/\sqrt{N}}, \quad s_d^2 = \frac{1}{N-1} \sum_{t=1}^N (x_{d,t} - \bar{x}_d)^2, \quad p_d = 2 \Pr(T_{756} \geq |t_d|).$$

3 Results

Table 1. Monthly excess returns by operating-profitability decile

OP decile	Mean (%)	Std. dev. (%)	t -statistic	p -value
1 (lowest)	0.4394	6.5959	1.8327	0.0672363
2	0.5142	5.3723	2.6336	0.0086206
3	0.4696	4.9263	2.6228	0.0088974
4	0.5619	4.6365	3.3345	0.0008961
5	0.6659	4.6271	3.9593	0.0000823
6	0.5924	4.5478	3.5837	0.0003605
7	0.5573	4.5699	3.3554	0.0008320
8	0.7272	4.6130	4.3375	0.0000164
9	0.7089	4.5306	4.3048	0.0000189
10 (highest)	0.6910	4.6210	4.1143	0.0000431

Notes: $N = 757$ for every decile. Means are arithmetic monthly averages; standard deviations use $N - 1$. Tests are two-sided with 756 degrees of freedom. P-values are unadjusted. Source: Kenneth French Data Library.

Main finding. Nine of ten deciles have positive mean excess returns significant at 5%. The lowest-profitability decile does not reject the zero-mean null ($p = 0.0672$).

4 Interpretation

Mean excess returns range from 0.4394% to 0.7272% per month. Deciles 2–10 reject the zero-mean null at both 5% and 1%. Decile 1 is significant at 10%, but not at 5% ($t = 1.8327$).

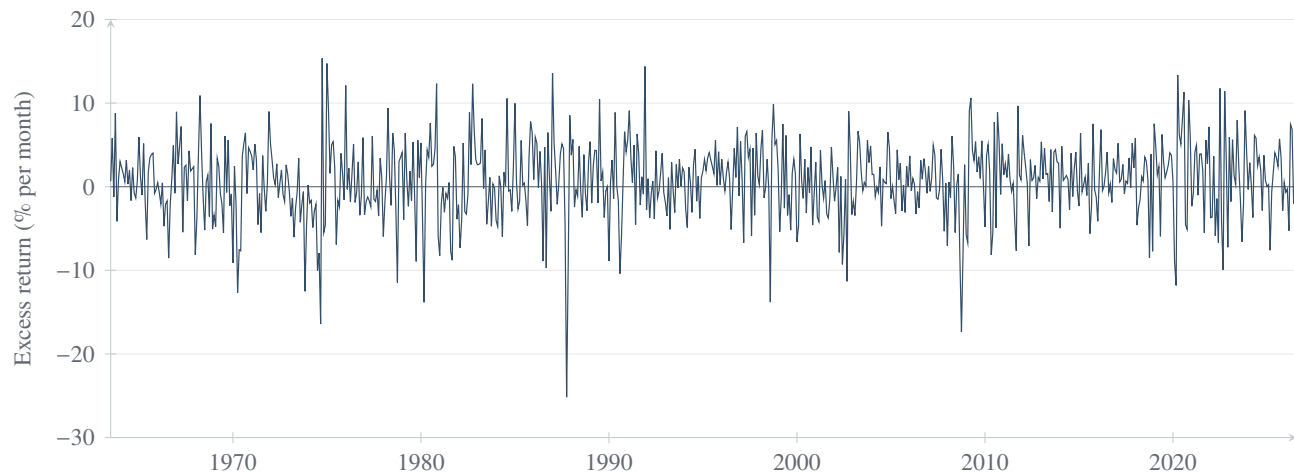
Higher-profitability portfolios generally have larger mean excess returns, although the pattern is not monotonic. Decile 8 has the highest mean (0.7272%); decile 10 averages 0.6910%. The lowest-profitability portfolio is also the most volatile: its monthly standard deviation is 6.5959%, compared with 4.6210% for decile 10.

The difference between the decile-10 and decile-1 sample means is approximately 0.252 percentage points per month. Individual portfolio tests do not establish the statistical significance of this difference. Standard errors do not adjust for serial correlation, and p-values are not corrected for multiple testing. These results do not establish risk-adjusted abnormal performance or a causal effect of profitability.

5 Monthly return variation

Figure 1. Highest-profitability decile: monthly excess return

Value-weighted OP decile 10 / July 1963–July 2026



Source: Kenneth French Data Library; portfolio return minus RF. The figure uses the same monthly observations as the Stata analysis.

Despite its positive sample mean, decile 10 experiences substantial month-to-month fluctuations. Its largest negative excess return occurs in October 1987 (−25.17%). The figure shows monthly returns, not cumulative investment performance.

Sources and reproducibility

Kenneth French Data Library; operating-profitability portfolio definitions. RF uses Ibbotson one-month Treasury bill data through May 2024 and the ICE BofA US 1-Month Treasury Bill Index thereafter. The library's current CIZ-based research return series is used.

The week-02 folder includes frozen source files, checksums, code and outputs. From the repository root in Stata, run `do run-week-02.do`. The appendix reproduces the analysis do-file. The report's editable LaTeX source is in `week-02/report/`.

APPENDIX A

Stata code

Data preparation and monthly excess returns

```

1  version 19.0
2  clear all
3  set more off
4  * Run from week-02, or use run-week-02.do from the repository root.
5  * Frozen July 2026 data vintage.
6  capture mkdir output
7  capture log close assignment1
8  log using "output/assignment1.log", text replace name(assignment1)
9
10 * 1. Downloaded from Kenneth French's Data Library on 15 September 2026.
11 * Raw ZIPs and CSVs are preserved in data/raw for reproducibility.
12 * Original URLs:
13 * https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/Portfolios_Formed_on_OP_CSV.zip
14 * https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/F-F_Research_Data_Factors_CSV.zip
15 * Row ranges select only the monthly blocks in this frozen vintage.
16 * OP columns 10-19 are the ten value-weighted decile portfolios.
17 import delimited using "data/raw/Portfolios_Formed_on_OP.csv", ///
18     rowrange(26:782) varnames(nonames) clear
19 rename v1 yyyymm
20 forvalues d = 1/10 {
21     local c = 'd' + 9
22     rename v'c' ret'd'
23 }
24 keep yyyymm ret1-ret10
25 isid yyyymm
26 tempfile portfolios
27 save 'portfolios'
28
29 * Monthly RF is column 5 of the Fama/French three-factor file.
30 import delimited using "data/raw/F-F_Research_Data_Factors.csv", ///
31     rowrange(6:1206) varnames(nonames) clear
32 rename v1 yyyymm
33 rename v5 rf
34 keep yyyymm rf
35 isid yyyymm
36 merge 1:1 yyyymm using 'portfolios'
37 assert _merge != 2
38 keep if _merge == 3
39 drop _merge
40 generate month = ym(floor(yyyymm/100), mod(yyyymm,100))
41 format month %tm
42 sort month
43 tsset month
44 assert month == month[_n-1] + 1 if _n > 1
45 assert _N == 757
46 assert month[1] == ym(1963,7) & month[_N] == ym(2026,7)
47
48 * 2. Returns and RF are in percent per month; retain these units.
49 foreach v of varlist ret1-ret10 rf {
50     replace 'v' = . if inlist('v', -99.99, -999)
51     assert !missing('v')
52 }
53 forvalues d = 1/10 {
54     generate double excess'd' = ret'd' - rf
55     label variable excess'd' "OP decile 'd': excess return (%)"
56 }
57 label variable month "Month"
58 label variable rf "One-month risk-free return (%)"
59 label data "Value-weighted OP deciles: July 1963-July 2026"
60 save "output/op_excess_returns.dta", replace

```

APPENDIX A / CONTINUED

Stata code

Hypothesis tests and Stata figure export

```

61 export delimited using "output/op_excess_returns.csv", replace
62
63 * 3. One-sample, two-sided t tests:  $H_0: E[\text{excess\_d}] = 0$ .
64 *  $t = \text{mean} / (\text{sample SD} / \sqrt{N})$ ;  $p = 2 * \text{ttail}(N-1, \text{abs}(t))$ .
65 tempfile statistics
66 tempname results
67 postfile 'results' byte decile int n double mean sd t p ///
68     using 'statistics', replace
69 forvalues d = 1/10 {
70     quietly ttest excess'd' == 0
71     post 'results' ('d') (r(N_1)) (r(mu_1)) (r(sd_1)) (r(t)) (r(p))
72 }
73 postclose 'results'
74 preserve
75 use 'statistics', clear
76 format mean sd t %9.4f
77 format p %10.7f
78 list, noobs separator(0)
79 save "output/summary_statistics.dta", replace
80 export delimited using "output/summary_statistics.csv", replace
81 restore
82
83 * 4. Time-series figure: highest operating-profitability decile.
84 twoway line excess10 month, ///
85     title("Highest-profitability decile: monthly excess return") ///
86     subtitle("Value-weighted OP decile 10 | July 1963-July 2026") ///
87     ytitle("Excess return (% per month)") xtitle("Year") ///
88     xlabel(0(120)720, format(%tmCCYY)) ///
89     yline(0, lcolor(gs9) lpattern(dash)) ///
90     lcolor(navy) lwidth(thin) graphregion(color(white)) ///
91     note("Source: Kenneth French Data Library; portfolio return minus RF.") ///
92     name(op_decile10, replace)
93 graph save "output/op_decile10.gph", replace
94 graph export "output/op_decile10.png", width(2400) replace
95 log close assignment1

```